$$\mathbf{F} = q(\mathbf{E} + \mathbf{v} \times \mathbf{B}) = m\mathbf{a} = m\frac{\partial \mathbf{v}}{\partial t} \qquad \nabla \cdot \mathbf{J} + \frac{\partial \rho}{\partial t} = 0$$

$$\oint_S \mathbf{D}\cdot\mathbf{dS} = \int_V \rho \, dV \qquad \nabla \cdot \mathbf{D} = \rho$$

$$\oint_S \mathbf{B} \cdot \mathbf{dS} = 0 \qquad \nabla \cdot \mathbf{B} = 0$$

$$\oint_C \mathbf{H}\cdot\mathbf{dl} = \int_S \mathbf{J} \cdot \mathbf{dS} + \frac{d}{dt}\int_S \mathbf{D} \cdot \mathbf{dS} \qquad \nabla \times \mathbf{H} = \mathbf{J} + \frac{\partial \mathbf{D}}{\partial t}$$

$$\oint_C \mathbf{E} \cdot \mathbf{dl} = -\frac{d}{dt}\int_S \mathbf{B} \cdot \mathbf{dS} \qquad \nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$$

$$\hat{n} \times (\mathbf{E}^+ - \mathbf{E}^-) = 0 \qquad \hat{n} \cdot (\mathbf{D}^+ - \mathbf{D}^-) = \rho_s$$

$$\hat{n} \cdot (\mathbf{B}^+ - \mathbf{B}^-) = 0 \qquad \hat{n} \times (\mathbf{H}^+ - \mathbf{H}^-) = \mathbf{J_s} \qquad \mathbf{H}^+ - \mathbf{H}^- = \mathbf{J_s} \times \hat{n}$$

**For static conditions:** $\nabla \times \mathbf{E} = 0$ $\qquad \mathbf{E} = -\nabla V$ $\qquad V_A - V_B = \int_A^B \mathbf{E} \cdot \mathbf{dl}$

$$\oint_C \mathbf{E} \cdot \mathbf{dl} = 0 \qquad \nabla^2 V = -\rho/\epsilon \qquad C = Q/V$$

## SOME STATIC E FIELDS (IN A VACUUM)

- point charge $Q$ [C] at the origin: $\mathbf{E} = \hat{r}\frac{Q}{4\pi r^2 \epsilon_0}$
- line charge $\rho_l$ [C/m] along the $z$-axis: $\mathbf{E} = \hat{r}\frac{\rho_l}{2\pi r \epsilon_0}$
- surface charge $\rho_s$ [C/m$^2$] in the $z = 0$ plane: $\mathbf{E} = \pm\hat{z}\frac{\rho_s}{2\epsilon_0}$ (for $z \gtrless 0$)
- slab charge $\rho$ [C/m$^3$] at $-\frac{W}{2} \le z \le \frac{W}{2}$ : $\mathbf{E} = \begin{cases} -\hat{z}\frac{\rho W}{\epsilon_0 2} & z \le -\frac{W}{2} \\ \hat{z}\frac{\rho}{\epsilon_0} z & -\frac{W}{2} \le z \le \frac{W}{2} \\ \hat{z}\frac{\rho W}{\epsilon_0 2} & z \ge \frac{W}{2} \end{cases}$
- cylindrical charge $\rho$ [C/m$^3$] of radius $R$ along the $z$-axis: $\mathbf{E} = \begin{cases} \hat{r}\frac{\rho R^2}{2\epsilon_0 r} & r \ge R \\ \hat{r}\frac{\rho}{2\epsilon_0} r & r \le R \end{cases}$

## SOME STATIC B FIELDS (IN A VACUUM)

- line current $I$ [A] along the $z$-axis: $\mathbf{B} = \hat{\phi}\frac{\mu_0 I}{2\pi r}$
- surface current $\mathbf{J_s} = J_s\hat{x}$ [A/m] at the $z = 0$ plane: $\mathbf{B} = \mp\hat{y}\frac{\mu_0 J_s}{2}$ (for $z \gtrless 0$)
- slab current $\mathbf{J} = J\hat{x}$ [A/m$^2$] at $-\frac{W}{2} \le z \le \frac{W}{2}$ : $\mathbf{B} = \begin{cases} -\hat{y}\ \mu_0 J\frac{W}{2} & z \ge \frac{W}{2} \\ \hat{y}\ \mu_0 J(\text{-}z) & -\frac{W}{2} \le z \le \frac{W}{2} \\ \hat{y}\ \mu_0 J\frac{W}{2} & z \le -\frac{W}{2} \end{cases}$

## E AND B FIELDS IN MATERIAL MEDIA

$$\mathbf{D} = \epsilon_0\mathbf{E} + \mathbf{P} = \epsilon\mathbf{E} \qquad \mathbf{H} = \frac{\mathbf{B}}{\mu_\mathbf{0}} - \mathbf{M} = \frac{\mathbf{B}}{\mu}$$

$$\mathbf{J_{free}} = \sigma\mathbf{E} \qquad \rho_{bound} = -\nabla\cdot\mathbf{P} \qquad \mathbf{J_{bound}} = \nabla\times\mathbf{M} + \frac{d\mathbf{P}}{dt}$$

$$\mathbf{P} = \epsilon_0\chi_e\mathbf{E} \qquad \mathbf{M} = \chi_m\mathbf{H}$$

$$\epsilon = \epsilon_0\epsilon_r = \epsilon_0(1+\chi_e)\ [\mathrm{F/m}] \qquad \mu = \mu_0\mu_r = \mu_0(1+\chi_m)\ [\mathrm{H/m}]$$

## EM WAVES (IN FREE SPACE)

for $\mathbf{E} = E_x(z,t)\hat{x}$ generated by source $\mathbf{J_s} = -J_s(t)\hat{x}$ at $z=0$: $\frac{\partial^2 E_x}{\partial z^2} = \mu\epsilon\frac{\partial^2 E_x}{\partial t^2}$

$$E_x(z,t) = \tfrac{\eta}{2}J_s(t\mp\tfrac{z}{v_p}) \ \text{ (for } z\gtrless 0) \qquad H_y(z,t) = \pm\tfrac{1}{2}J_s(t\mp\tfrac{z}{v_p}) \ \text{ (for } z\gtrless 0)$$

$$v_p = \tfrac{1}{\sqrt{\mu\epsilon}} \qquad c = \tfrac{1}{\sqrt{\mu_0\epsilon_0}} = 3\times10^8\ \mathrm{m/s} \qquad \eta = \sqrt{\tfrac{\mu}{\epsilon}} \qquad \eta_0 = \sqrt{\tfrac{\mu_0}{\epsilon_0}} = 120\pi$$

for source $\mathbf{J_s} = -\cos(\omega t+\phi)\hat{x}$ at $z=0$: $E_x(z,t) = \frac{\eta}{2}\cos(\omega t\mp\beta z+\phi)$ (for $z\gtrless 0$)

$$v_p = \tfrac{\omega}{\beta} = \lambda f \qquad \beta = \tfrac{2\pi}{\lambda} \qquad \omega = \tfrac{2\pi}{T} = 2\pi f$$